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Subject: Machine Learning

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EXPERIMENT 7

AIM: To implement SVM Classifier

Code:



```
from google.colab import files
uploaded = files.upload()

import pandas as pd
import numpy as np

df = pd.read_csv(list(uploaded.keys())[0])

print("Dataset Preview:")
print(df.head())

X = df.drop('Monthly Mean Total Sunspot Number', axis=1)
y = df['Monthly Mean Total Sunspot Number']

X = pd.get_dummies(X)

from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```



```
from sklearn.svm import SVR
from sklearn.metrics import mean_squared_error, r2_score

model = SVR(kernel='linear') # You can try 'rbf', 'poly' also
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("\nMean Squared Error:", mean_squared_error(y_test, y_pred))
print("\nR-squared Score:", r2_score(y_test, y_pred))
```

```
** Choose files archive.zip
archive.zip(application/x-zip-compressed) - 22972 bytes, last modified: 27/03/2026 - 100% done
Saving archive.zip to archive (2).zip
Dataset Preview:
   Unnamed: 0    Date  Monthly Mean Total Sunspot Number
0           0  1749-01-31                                96.7
1           1  1749-02-28                               104.3
2           2  1749-03-31                               116.7
3           3  1749-04-30                                92.8
4           4  1749-05-31                               141.7

Mean Squared Error: 4479.4286450700165
R-squared Score: -6.533383274942572e-05
```

Conclusion:

Hence we implemented svm classifier where svm stands for support vector machine and it helps to separate the data so if we have red and blue dots it will separate it using a line called hyperplane.